

# METRIC-AWARE CONTRASTIVE REGULARIZATION OF OBSERVATIONS

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## ABSTRACT

Model-based reinforcement learning (MBRL) agents struggle to master long-horizon tasks with extremely sparse rewards. A fundamental bottleneck, exemplified by the Atari *Seaquest* environment, extends beyond a simple representation paradox. Even if an agent successfully encodes minute, static, yet critical sub-goal metrics such as UI counters into the latent space, modern encoding-based world models fail to map the causal dynamics underlying these metric variations to the environment’s long-horizon reward chain. To overcome this, we introduce `ImportantInfoHashLoss`, a targeted representation constraint leveraging historical samples retrieved from the entire replay buffer—including both a single seed trajectory and online experiences. Rather than relying on computationally heavy full-frame pixel reconstruction or settling for mere visual retention, our method forces a value-driven structuring of functionally distinct states through a metric-aware contrastive objective. By hashing latent representations for efficient historical retrieval, our method explicitly forces a geometric separation between embeddings that are visually similar but functionally distinct—effectively breaking the encoder’s smoothing bias over static UI variables. Furthermore, we propose a Reward Concatenation mechanism that injects past rewards directly into the input of a Mamba3 sequence model, enriching the latent transition dynamics. Empirical evaluations demonstrate that our framework successfully aligns the agent’s internal value estimation with environmental progress, entirely without the use of artificial reward shaping or predefined milestones. Without any manual reward shaping, the agent’s expected value function scales proportionally and step-wise as the number of collected divers increases, providing a computationally efficient paradigm for solving complex, sparse-reward challenges. Our code is available at [https://github.com/wrrte/graduate\\_research.git](https://github.com/wrrte/graduate_research.git)

## 1 INTRODUCTION

Model-based reinforcement learning (MBRL) has achieved remarkable success by learning compressed latent dynamics of environments, enabling efficient planning and policy optimization (?). However, mastering long-horizon tasks with extremely sparse rewards remains a formidable challenge. A quintessential example is the Atari *Seaquest* environment, where an agent must navigate complex dynamics and collect six divers before surfacing to secure a massive sparse reward—which far outweighs the trivial dense rewards of simply shooting enemies. Solving such tasks requires the agent not only to execute extended sequences of actions but also to persistently retain task-critical metrics—such as the exact count of rescued divers—to correctly map the causal relationship between its internal progress and the ultimate reward over thousands of environmental steps.

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This challenge exposes a critical representation paradox in contemporary encoding-based world models (e.g., ???). Theoretically, these architectures are designed to preserve all task-relevant information while discarding irrelevant environmental noise. In practice, however, their representation learning objectives exhibit a strong inductive bias toward either high-magnitude visual dynamics—such as moving enemies and expanding explosions—or any visual features directly coupled with immediate rewards. Consequently, without artificial reward shaping to explicitly anchor them, the encoder inherently prioritizes these readily learnable signals while discarding minute, static, and unrewarded sub-goal metrics as background noise.

A more critical limitation is that mere pixel-level reconstruction or representation retention is functionally insufficient for solving temporal credit assignment. Even if a world model distinctly encodes static sub-goal metrics visually, the policy network completely collapses if it fails to learn the causal rule that variations in this metric arise from specific environmental interactions (e.g., colliding with a submarine). Unless functionally distinct states are geometrically and explicitly separated within the value space, the agent treats them as actionable noise, failing to establish a causal link with the delayed sparse reward.

While recent diffusion-based world models (e.g., EDELIN; ?) have achieved high scores in *Seaquest* through maximal pixel-level reconstruction, they incur prohibitive computational costs. Although they employ State Space Models (SSMs) to theoretically support infinite context lengths via their recurrent state representations, the heavy computational burden of iterative pixel generation practically restricts their scalability for extremely long-horizon tasks.

To resolve this efficiency-fidelity dilemma, we propose a unified model-based framework that enforces the selective retention of critical static variables without relying on full-frame pixel reconstruction. Given the prohibitive exploration space of long-horizon sparse-reward tasks, we initialize the replay buffer with a single seed trajectory. This minimal prior ensures the agent is aware of the delayed sparse reward and its underlying temporal structure, without prescribing strict behavioral cloning. Building upon this, we introduce `ImportantInfoHashLoss`, a targeted representation learning constraint leveraging historical samples retrieved from the entire replay buffer—encompassing both the single seed trajectory and ongoing online experiences. Rather than forcing the model to reconstruct every pixel, this loss explicitly etches these minute sub-goal metrics into the latent space via a metric-aware contrastive constraint.

By clustering historically similar latent states using a fast Random Projection-based SimHash algorithm, the loss explicitly penalizes the cosine similarity of representations that share visual features but differ in their underlying state values. Consequently, this precise disentanglement provides a reliable foundation for accurate downstream temporal metrics, such as value or TD-error. This elegantly pulls functionally distinct states apart in the latent space. Furthermore, we propose an input-level Reward Concatenation mechanism that injects past two-hot encoded reward signals directly into a Mamba3 sequence backbone alongside latent states and actions, significantly enriching the transition dynamics under sparse-reward regimes.

The main contributions of this paper are summarized as follows:

- Motivated by the empirical insight that achieving perfect pixel-level representation does not naturally translate to successful policy execution, we systematically isolate the fundamental disconnect between passive encoding and the learning of causal dynamics in modern MBRL. We provide direct evidence that the inability to structurally separate discrete task counters within the value space—rather than the mere failure to ‘memorize’ them—is the core culprit behind credit assignment failures in sparse-reward tasks.
- We introduce `ImportantInfoHashLoss`, a targeted representation constraint leveraging historical samples retrieved from the entire replay buffer—encompassing both the single seed trajectory and online interaction data—to preserve critical but static visual variables in the latent space.
- We propose an enriched transition architecture that pairs input-level Reward Concatenation with a Mamba3 backbone to trace long-horizon context without information decay.
- We empirically demonstrate that our framework aligns the agent’s internal value estimation with true environmental progress, entirely without the use of artificial reward shaping or predefined milestones. Without any manual reward shaping, the expected value function

scales proportionally and step-wise as the number of collected divers increases from zero to six.

## 2 RELATED WORK AND MOTIVATION

**World Models and Representation Limits.** Recent advancements in model-based reinforcement learning (MBRL), largely driven by the Dreamer family (?) and DRAMA, leverage latent dynamics to achieve remarkable sample efficiency. However, these models inherently apply a lossy compression bottleneck optimized to filter out perceptual noise. In visually complex but reward-sparse environments like *Atari Seaquest*, this inductive bias aggressively discards static, non-moving UI elements—such as the accumulated diver count—because they exhibit negligible spatio-temporal variance compared to moving background sprites. Consequently, the agent’s latent state loses the exact information required to establish long-term credit assignment.

**The Diffusion Proof-of-Concept: EDELINE.** The hypothesis that standard encoders inadvertently discard crucial task variables was implicitly corroborated by EDELINE (?), which achieved state-of-the-art performance (2,140.1 in *Seaquest* and 4,224.5 in *Asterix*) on the Atari 100k benchmark among non-lookahead search methods. EDELINE bypasses the representation paradox by employing diffusion-based generative models to enforce pixel-perfect reconstruction of the entire environmental frame. While full pixel-level reconstruction theoretically offers an exhaustive and ideal representation of the environment, its practical deployment introduces a prohibitive computational bottleneck. Operating diffusion models over extended long-horizon trajectories is heavily resource-intensive, where the immense computational overhead required for iterative pixel generation practically limits its scalability in environments requiring thousands of sequential steps.

**Long-Term Memory and a Single Seed Trajectory.** To handle extended contexts without full-frame reconstruction, recent architectures incorporate advanced sequence modeling and external memory, such as ELMUR (?) and MemoryVLA (?). While these state-space models (SSMs) and memory banks excel at long-horizon context retention, they still rely on the underlying visual encoder to extract the correct features initially. Furthermore, accelerating representation learning in extremely sparse environments frequently requires leveraging a single seed trajectory to bootstrap the correct transitions.

**Motivation.** The juxtaposition of EDELINE’s high-fidelity success and the computational redundancy of full-frame reconstruction directly motivates our approach. If the primary requirement for solving *Seaquest* is the unbroken retention of specific static sub-goal metrics, full-frame diffusion is an over-parameterized solution. Furthermore, simply replacing a recurrent backbone with advanced sequence models like Mamba3 is insufficient if the visual encoder has already discarded the vital UI information, or if the transition dynamics lack a direct temporal mechanism to correlate these variables with delayed rewards.

Therefore, we propose a unified architecture that grafts selective semantic retention and dynamic enrichment directly into an efficient MBRL framework. By computing the `ImportantInfoHashLoss` through historical samples retrieved from a comprehensive replay buffer—seeded by a single seed trajectory and continuously expanded by online experiences—we force the latent space to anchor the static UI variables. Concurrently, by concatenating past rewards directly into the Mamba3 input, we explicitly enrich the sequence model’s capacity to correlate these preserved static states with the delayed sparse rewards. This unified strategy achieves the causal grounding of EDELINE at a fraction of the computational cost, providing a scalable pathway for memory-augmented MBRL.

## 3 PROPOSED METHOD

### 3.1 OVERALL SYSTEM ARCHITECTURE

To fundamentally solve the representation paradox and credit assignment failure in long-horizon, sparse-reward tasks, we introduce a unified model-based framework that augments latent sequence modeling with selective semantic grounding. Building upon the DRAMA architecture, our framework introduces three targeted modifications designed to overcome the representation

paradox: an upgraded Mamba3-based sequence backbone, an input-level Reward Concatenation mechanism, and the `ImportantInfoHashLoss` regularization. Importantly, these components are strictly complementary, addressing opposite ends of the reinforcement learning spectrum. While Reward Concatenation stabilizes short-term reactive behaviors in dense-reward regimes, the `ImportantInfoHashLoss` tackles the challenges of long-horizon sparse rewards by anchoring strategic task progress.

The overall operational transition of our world model at timestep  $t$  proceeds as follows. Given an environmental observation  $o_t$ , a convolutional encoder extracts the visual features to generate a stochastic latent state  $z_t \sim q(z_t|o_t)$ . Unlike conventional world models that unfold the deterministic recurrent state solely based on the latent state and the preceding action, our architecture enriches the transition dynamics by directly concatenating the past reward signal. Specifically, the scalar reward  $r_{t-1}$  is symlog-transformed, encoded via a two-hot representation, and concatenated alongside  $z_t$  and the action  $a_t$ . This unified vector serves as the complete operational context for the Mamba3 sequence model:

$$I_t = [z_t \parallel a_t \parallel \text{TwoHot}(\text{Symlog}(r_{t-1}))] \quad (1)$$

The Mamba3 backbone processes the sequence of enriched inputs  $I_{1:t}$  in linear time to generate the deterministic hidden state  $h_t = \text{Mamba3}(I_t, h_{t-1})$ . The combined state  $s_t = (z_t, h_t)$  represents the comprehensive latent belief of the environment, which is subsequently utilized by the policy network (Actor) and the value network (Critic).

Crucially, to ensure that the visual encoder does not discard minute, static UI assets essential for tracing long-term task progression, the entire world model is optimized under a targeted representation constraint. During the training phase, batches of trajectories are sampled uniformly from the comprehensive replay buffer, which is pre-populated with a single seed trajectory alongside the agent’s ongoing online interaction data. It is crucial to note that the primary objective of introducing this single seed trajectory is not to accelerate initial learning via imitation. Rather, it serves to reveal the existence of the immensely delayed sparse reward to the agent and to provide a foundational example of the valid sequential process required to attain it. In addition to standard reconstruction and transition losses, the visual representation is regularized via the `ImportantInfoHashLoss`. This mechanism persistently queries offline and online experiences to identify visually aliased states, forcing a structural separation in the latent coordinates of  $z_t$  whenever their underlying sub-goal metrics diverge.

### 3.2 METRIC-AWARE CONTRASTIVE REGULARIZATION VIA `IMPORTANTINFOHASHLOSS`

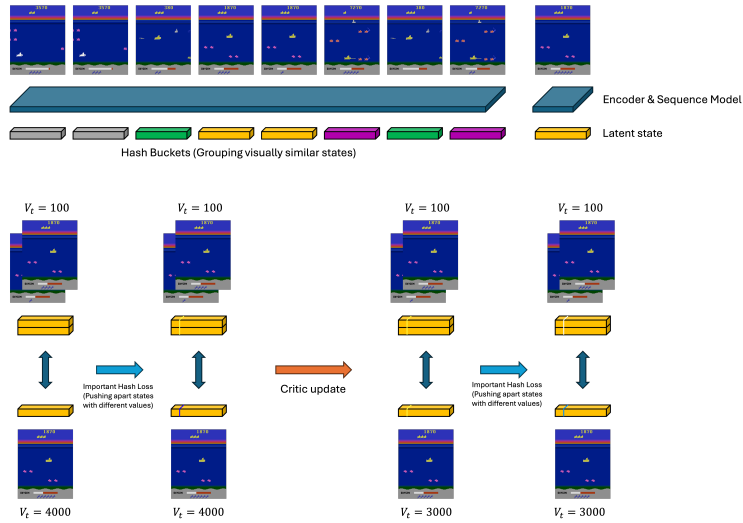


Figure 1: Conceptual illustration of `ImportantInfoHashLoss`. Visually aliased states are initially mapped to the same hash bucket due to the encoder’s smoothing bias. Our value-aware contrastive objective explicitly forces a geometric separation between these functionally distinct embeddings in the latent space based on their internal value divergence.

The fundamental cause of the representation paradox is visual aliasing: the convolutional encoder’s tendency to map functionally distinct states into highly similar, collapsed latent representations due to the negligible pixel variance of static UI elements. To counteract this smoothing bias efficiently, we introduce `ImportantInfoHashLoss`, a value-aware contrastive regularization that resolves latent visual aliasing without relying on computationally heavy full-frame reconstruction.

**Random Projection Hashing for  $\mathcal{O}(1)$  Retrieval.** To efficiently identify visually aliased states across the comprehensive replay buffer, we utilize a fast Random Projection-based SimHash algorithm. A continuous latent representation  $z_t$  is projected into a  $B$ -bit discrete hash key  $k \in \mathbb{Z}$  via a fixed Gaussian random projection matrix  $W_{proj} \sim \mathcal{N}(0, 1)$ :

$$k = \sum_{i=1}^B 2^{i-1} \mathbb{I}(z_t \cdot W_{proj} > 0) \quad (2)$$

This operation preserves the cosine similarity of the latent space. Consequently, the encoder inadvertently assigns visually identical states to the same hash bucket within the memory bank, enabling  $\mathcal{O}(1)$  retrieval time to capture occurrences of visual aliasing without exhaustive pairwise comparisons.

**Dynamic Value-Weighted Contrastive Penalty via EMA.** For a given state  $z_t$  with hash key  $k$ , the module retrieves historical states  $z_{past}$  residing in the same bucket. Because they share the same hash, the encoder perceives them as visually identical. To force a geometric separation based on their actual semantic differences, we calculate the absolute value divergence  $\Delta m = |m_t - m_{past}|$ , where  $m$  represents the internal state-value estimates predicted by the network.

In long-horizon reinforcement learning, the scale of internal value estimates  $m$  grows exponentially as the agent discovers delayed sparse rewards, creating a highly non-stationary training regime. Initially, utilizing a standard Welford’s algorithm to compute cumulative running statistics caused a severe over-triggering bug; the historical average became anchored to the near-zero values of the early exploration phase, causing normal mid-to-late game value variations to be systematically misclassified as extreme outliers. To rectify this and ensure the loss acts as a precise pulse exclusively for critical task events,  $\Delta m$  is dynamically normalized by an adaptive bound ( $\max |m|$ ) computed via an Exponential Moving Average (EMA). By prioritizing recent statistical trends, the EMA-based normalization fluidly adapts to the rapidly escalating value scales.

The contrastive objective then penalizes the cosine similarity between the current and past pre-sampled latent logits ( $x_t$ ), explicitly weighted by this EMA-normalized divergence:

$$\mathcal{L}_{hash} = \frac{1}{N} \sum \left( \frac{|m_t - m_{past}|}{\text{EMA}(\max |m|)} \cdot \max(0, \cos(x_t, x_{past})) \right) \quad (3)$$

(Note: To prevent representation collapse and stabilize the optimization process, standard stop-gradient operations are implicitly applied to both the value divergence multiplier and the historical representations.)

The formulation of this value-weighted penalty introduces a critical regularizing property: when two visually aliased states possess a significant disparity in their expected future returns ( $\Delta m > 0$ ), the loss heavily penalizes the encoder, driving a structural separation in the latent topology. Conversely, if the states share identical internal values ( $\Delta m = 0$ ), the penalty gracefully vanishes to zero. This inherently permits representation collapse for value-equivalent states, preventing the encoder from wasting its expressive capacity on task-irrelevant visual variances (e.g., background water ripples or trivial pixel noise).

### 3.3 DYNAMICS ENRICHMENT VIA INPUT-LEVEL REWARD CONCATENATION

Standard recurrent world models typically update their sequence transition dynamics using only the subsequent latent state and the executed action, following the recurrence relation  $h_t = f(h_{t-1}, z_t, a_t)$ . In an environment characterized by extremely delayed and sparse rewards, this formulation isolates the latent transition dynamics from historical reward contexts, forcing the value network to map sparse targets from decoupled hidden features.

To rectify this limitation, we propose an input-level Reward Concatenation mechanism that directly enriches the transition dynamics inside the sequence model. Prior to being fed into the temporal

backbone, the scalar reward from the previous timestep  $r_{t-1}$  is first compressed via a symlog transformation to stabilize its variance, and subsequently transformed into a robust representation using a two-hot encoding scheme, denoted as  $\text{TwoHot}(\text{Symlog}(r_{t-1})) \in \mathbb{R}^d$ . This dual-encoding process prevents numerical underflow and scales the sparse reward signal to match the magnitude of the latent features seamlessly.

The unified context vector  $I_t$  is constructed via a direct concatenation operation in the feature dimension:

$$I_t = \text{Concat}(z_t, a_t, \text{TwoHot}(\text{Symlog}(r_{t-1}))) \quad (4)$$

By incorporating  $\text{TwoHot}(\text{Symlog}(r_{t-1}))$  directly into  $I_t$ , the world model is explicitly forced to contextualize its current state transition based on whether an immediate reward was just secured. This architectural modification provides a direct temporal bridge for reactive events—such as shooting enemies in *Seaquest* or collecting clues in *Private Eye*—ensuring that the sequence model remains highly responsive to short-term, high-frequency environmental feedback.

### 3.4 LONG-HORIZON SEQUENCE MODELING VIA ADVANCED MAMBA-3 BACKBONE

Once the dynamics-enriched context vector  $I_t$  is constructed, the sequence model must propagate this consolidated multi-modal representation across extended temporal horizons. While traditional world models utilize recurrent loops to maintain a deterministic hidden state  $h_t = f(h_{t-1}, I_t)$ , they suffer from severe information decay over thousands of steps. To ensure robust credit assignment in long-horizon tasks, we employ an advanced Mamba-3 sequence backbone, which governs the temporal transition dynamics through a highly selective, structurally enhanced state-space formulation.

Unlike classic state-space models that apply fixed transition matrices or basic input-dependent discretizations, our Mamba-3 backbone incorporates explicit positional tracking and dynamic filtering. Operating on the unified context vector  $I_t$ , the continuous-time latent evolution is discretized via a step size  $\Delta_t = \text{Softplus}(\text{Linear}(I_t))$ . Crucially, Mamba-3 enhances the memory retention of critical events (e.g., reward spikes) by introducing a *Trap* gating mechanism ( $g_t$ ) and applying Rotary Position Embeddings (RoPE) and RMSNorm to the projection matrices. The sequence parameters are formalized as follows:

$$g_t = \sigma(\text{Linear}_{\text{trap}}(I_t)) \quad (5)$$

$$\tilde{B}_t = \text{RoPE}(\text{RMSNorm}(\text{Linear}_B(I_t))) \quad (6)$$

$$\bar{A}_t = \exp(\Delta_t A) \quad (7)$$

$$\bar{B}_t = (\Delta_t \tilde{B}_t) \odot g_t \quad (8)$$

where  $A$  represents a structured hardware-efficient matrix,  $\sigma$  is the sigmoid activation, and  $\odot$  denotes element-wise multiplication. The recurrent hidden state  $h_t$ , which compresses the entire historical trajectory of states, actions, and rewards, evolves according to the linear recurrence relation:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t I_t \quad (9)$$

This mathematically rigorous formulation directly addresses the long-horizon bottleneck. Traditional recurrent baselines like Long Short-Term Memory (LSTM) networks rely on strictly partitioned, independently learned gates to manage information routing, which often fails to adapt fluidly to heterogeneous inputs. In contrast, our Mamba-3 backbone harmoniously orchestrates all multi-modal contexts—states, actions, and historical rewards—within a unified stream. By exploiting its data-dependent selection mechanism combined with the specialized Trap gate  $g_t$ , the network fluidly determines what to retain or discard based on the exact environmental situation. This allows the world model to explicitly capture and "trap" rare but critical historical events—such as the reward spikes injected via our input concatenation—into the hidden state  $h_t$  without being prematurely overwritten by subsequent trivial background frames. Simultaneously, the application of RoPE on the  $B$  matrix ensures that the exact temporal distance between these trapped reward markers and past sub-goals is structurally preserved.

During training, this advanced linear recurrence is mapped onto a highly parallelized associative scan operation, bypassing the sequential computation bottleneck inherent in traditional RNNs and establishing an  $\mathcal{O}(T)$  linear time complexity. Consequently, the world model successfully maintains a high-fidelity temporal anchor over thousands of environmental steps without experiencing gradient

vanishing. This allows the downstream value and policy networks to perform precise long-term credit assignment, leveraging an uncorrupted history of state values and reactive feedback to escape sub-optimal local minima.

## 4 EXPERIMENTS AND RESULTS

### 4.1 EXPERIMENTAL SETUP

We evaluate our proposed unified framework on the Atari *Seaquest* environment, a benchmark notorious for its long-horizon challenges and extremely sparse reward structure. Our agent is built upon an advanced Mamba-3-based model-based reinforcement learning architecture that leverages long-horizon latent imagination, augmented with the proposed input-level Reward Concatenation and `ImportantInfoHashLoss` operating over historical samples retrieved from the entire replay buffer.

To ensure robust exploration and prevent initial distribution shifts, the replay buffer is initialized with a standard mixture of a single seed trajectory and random exploratory transitions. The policy subsequently interacts with the environment to accumulate online experiences.

While maximizing the final episodic score remains the ultimate objective, standard recurrent world models entirely fail to map the causal dynamics of the static diver UI. Therefore, to evaluate the fundamental stepping stones toward full-task mastery, our primary evaluation metric focuses on the *internal representation learning* and the *value alignment* of the agent during the unrewarded sub-goal sequence.

### 4.2 SEMANTIC GROUNDING OF UNREWARDED PROGRESS

The core hypothesis of this paper is that enforcing the retention of static UI elements allows the agent to intrinsically understand its progress toward a delayed sparse reward. To empirically validate this, we conduct a cross-sectional evaluation of the agent’s internal value function  $V(s_t)$ .

We sampled a diverse set of highly randomized environmental states (varying player positions, enemy submarine layouts, and oxygen levels), strictly categorized by the exact number of rescued divers currently held by the agent ( $n \in \{0, 1, \dots, 6\}$ ). For each category, we extracted multiple independent frames and computed the average predicted auxiliary value (`aux_value`) from the Critic network.

As demonstrated in Figure 2, the results show a striking step-wise monotonic increase in the expected value function directly correlated with the accumulated diver count. It is crucial to note that although the environment provides dense rewards for reactive combat (e.g., shooting fish or enemies) and a massive sparse reward for successfully surfacing with a full crew, the specific semantic transition of securing a diver yields exactly zero immediate external reward. Without any manual reward shaping or heuristic guidance, the agent autonomously maps the incremental changes in the minute, static UI counter to a proportionally escalating future return expectation. This effectively proves that the metric-aware contrastive mechanism of `ImportantInfoHashLoss` successfully broke the visual aliasing, forcing the latent embedding  $z_t$  to geometrically separate the static UI states, while the Mamba3 sequence model and Reward Concatenation maintained this distinct temporal context without information decay.

**Generalization to Highly Dynamic Environments: Asterix Benchmark.** To rigorously evaluate the robustness and versatility of MACRO beyond sparse-reward settings with static sub-goal metrics, we extend our evaluation to *Asterix*—a highly dynamic, dense-reward environment characterized by fast-moving, reactive objects. In this distinct setting, MACRO demonstrates remarkable generalization capabilities, achieving a competitive performance that closely approaches  $\Delta$ -IRIS, a sequence-based world model explicitly optimized for temporal frame differences. Furthermore, our framework successfully recovers approximately half the performance of the state-of-the-art diffusion-based model, EDELIN (4,224.5), while significantly outperforming standard encoding-based baselines such as DreamerV3. This strong empirical result underscores a crucial structural advantage: although MACRO introduces a targeted hash-based contrastive regularization to pre-

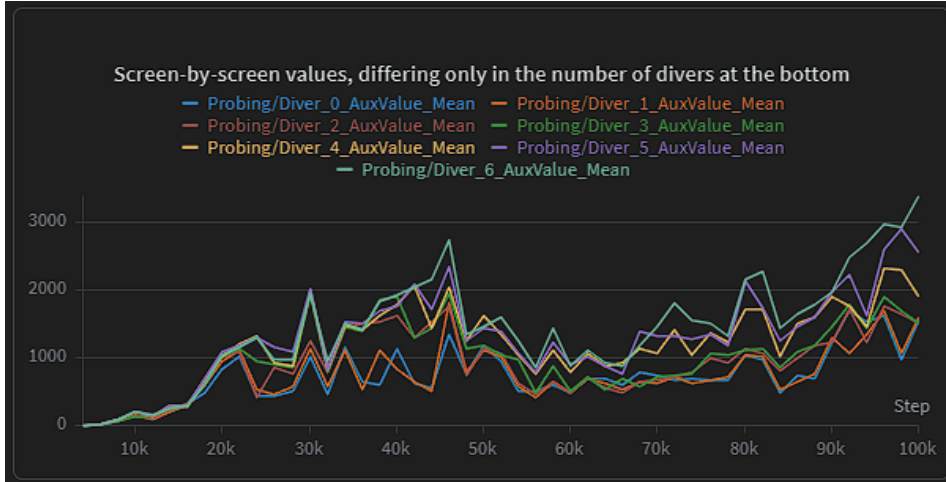


Figure 2: Overall step-wise scaling trend of the internal auxiliary value function corresponding to the exact number of rescued divers (0 to 6) under the MACRO framework.

serve minute, invariant static metrics, it does so without compromising the encoder’s capacity to compress and track high-magnitude, fluid visual dynamics.

This generalization highlights the intuitive synergy between our proposed modules across diverse task structures. In complex environments like *Seaquest*, accumulating dense rewards (e.g., shooting enemies) is a prerequisite for surviving long enough to achieve the sparse reward (e.g., surfacing with six divers). While Reward Concatenation directly accelerates the learning of these dense, immediate dynamics, the `ImportantInfoHashLoss` ensures that the overarching sparse goal is not lost amidst the intense local interactions. Consequently, their integration systematically improves the baseline by allowing dense reward tracking to act as a stepping stone toward delayed sparse objectives, without mutual interference.

#### 4.3 THE INSUFFICIENCY OF PURE REPRESENTATION: WHY STRUCTURING MATTERS

Under a rigorously stabilized baseline regime, we observed instances where the baseline encoder successfully retained the static diver UI in the latent space, overcoming the initial visual aliasing issue.

However, the most critical finding was the agent’s reaction to this retained information: despite successfully ‘seeing’ the exact diver count, the policy network exhibited severe behavioral collapse. The agent degenerated into suboptimal, defensive policies—hovering near the water surface, continuously discharging projectiles, and entirely abandoning the long-horizon rescue objective.

This empirical observation exposes a fundamental limitation in current MBRL architectures: mere pixel-level representation of a sparse metric is functionally insufficient. Even though the world model distinctly encodes the static sub-goal visual, it completely fails to learn the causal dynamics underlying it—specifically, the transitional rule that colliding with a diver object leads to its disappearance and a subsequent increment in the UI counter. Without external rewards guiding this exact semantic transition, the Actor treats the UI changes as actionable noise.

This disconnect firmly establishes the necessity of our `ImportantInfoHashLoss`. It demonstrates that deep reinforcement learning in sparse environments requires more than just passive representation (reconstruction); it inherently demands a metric-aware contrastive objective that structurally separates functionally distinct states based on their expected value, actively anchoring static visuals to the task’s causal structure.



## 5 CONCLUSION AND FUTURE WORK

### 5.1 CONCLUSION

In this paper, we investigated the representation paradox in model-based reinforcement learning, where modern encoding-based world models suffer from severe visual aliasing over minute, static sub-goal metrics in long-horizon, sparse-reward environments. To address this, we proposed a unified framework that combines a Mamba3 sequence backbone, an input-level Reward Concatenation mechanism, and `ImportantInfoHashLoss` leveraging historical samples retrieved from the entire replay buffer. Our empirical results on Atari *Seaquest* demonstrated that our metric-aware contrastive regularization successfully breaks the encoder’s smoothing bias. Without any manual reward shaping, our framework geometrically separates functionally distinct UI states within the latent space, allowing the internal value function to exhibit a clear upward scaling trend, effectively aligning with unrewarded progress metrics.

### 5.2 CURRENT LIMITATIONS AND FAILURE MODES: THE ARCHITECTURAL MEMORY BOTTLENECK

Despite the successful causal anchoring provided by `ImportantInfoHashLoss`, the agent still occasionally struggles to robustly execute the complete six-diver rescue trajectory. Through our stabilized evaluations, we identified that this remaining failure mode stems from an architectural memory bottleneck rather than a lack of control capacity (e.g., obstacle avoidance or oxygen management).

In the current MBRL framework, both non-causal spatial mixing and causal temporal dynamics are compressed into a single sequence model’s fixed-size recurrent state. Over the extreme temporal horizons required in *Seaquest* (often spanning thousands of steps), the continuous influx of high-magnitude visual noise inevitably degrades the sparse, exact recall of distant sub-goal events. For instance, the exact memory of a specific causal transition—such as collecting the fourth diver hundreds of steps prior—gradually decays within the lossy compression of the SSM backbone.

Because of this temporal degradation, the geometric separation of states achieved by our contrastive objective weakens in the exceptionally late stages of the episode ( $n \geq 4$ ). When these crucial temporal anchors fade, the agent experiences late-stage policy instability, occasionally losing track of its progress and failing to surface in time. This failure mode provides a critical insight: while `ImportantInfoHashLoss` resolves the representation paradox at the objective level, maintaining these structural anchors across infinite horizons requires an explicit architectural mechanism dedicated to exact episodic recall.

### 5.3 FUTURE TRAJECTORIES FOR ICLR 2027

While our framework successfully anchors the semantic representation of long-term progress at the objective level, fully mastering *Seaquest* requires addressing both numerical robustness and architectural memory bottlenecks. To achieve competitive full-task execution for ICLR 2027, our primary future trajectories focus on two orthogonal directions.

**Numerical Robustness and Precision Ablations.** Under our current stabilized setup, the baseline inadvertently succeeded in encoding the static UI, likely aided by the high-precision computational environment. However, under constrained precision policies (e.g., FP16 quantization or strict TF32 regimes), standard encoders are highly susceptible to severe representation cliffing. To rigorously validate the robustness of our `ImportantInfoHashLoss`, we plan to artificially induce visual aliasing by systematically degrading the underlying numerical precision. We hypothesize that our metric-aware contrastive objective will forcefully preserve the geometric separation of sub-goal metrics even under these extreme precision constraints, proving its hardware-agnostic necessity.

**Integrating the REMI Architecture for Episodic Recall.** To directly resolve the architectural memory degradation identified in the late-game stages, we plan to integrate the REMI (Remember to Imagine) factorized hybrid world model architecture. REMI explicitly disentangles non-causal spatial mixing, causal temporal dynamics, and exact episodic recall into distinct operations. Specifically, by incorporating REMI’s surprise-gated episodic memory, the agent can actively store and

cross-attend to exact latent snapshots of critical events (e.g., diver collection), triggered by high one-step NLL 'surprise' spikes that inherently align with our value divergence pulses.

We anticipate that pairing our `ImportantInfoHashLoss` (which structurally separates distinct functional states at the loss level) with the REMI backbone (which exactly recalls them across infinite horizons at the architectural level) will comprehensively bridge the gap between representation learning and long-horizon credit assignment.

## 6 APPENDIX

You may include other additional sections here.